Ex No: 1

16/7/2026

BASIC NETWORKING COMMANDS

Aim:

To implement basic Networking Commands in linux and Windows Operating system

BASIC NETWORKING COMMANDS:

=> arp -a :

ARP is short form of address resolution protocol.

=> hostname:

This is the simplest form of all TCP / IP commands.

=> ipconfig /all :

This displays detailed configuration information about connection including Router, Gateway, DNS, DHCP

```
PS C:\Users\DELL> arp -a

Interface: 192.168.56.1 --- 0x11
  Internet Address      Physical Address
  192.168.56.255        ff-ff-ff-ff-ff-ff
  224.0.0.2             01-00-5e-00-00-02
  224.0.0.22            01-00-5e-00-00-16
  224.0.0.251           01-00-5e-00-00-fb
  224.0.0.252           01-00-5e-00-00-fc
  224.0.0.253           01-00-5e-00-00-fd
  224.0.1.60            01-00-5e-00-01-3c
  239.0.0.8             01-00-5e-00-00-08
  239.192.152.143       01-00-5e-40-98-8f
  239.255.67.250        01-00-5e-7f-43-fa
  239.255.102.18        01-00-5e-7f-66-12
  239.255.255.250       01-00-5e-7f-ff-fa
```

```
C:\Users\DELL>nslookup
Default Server:  UnKnown
Address:  172.16.72.1
```

```
PS C:\Users\DELL> netstat -a

Active Connections

  Proto  Local Address
  TCP    0.0.0.0:135
  TCP    0.0.0.0:445
  TCP    0.0.0.0:3306
  TCP    0.0.0.0:5040
  TCP    0.0.0.0:33060
  TCP    0.0.0.0:49664
  TCP    0.0.0.0:49665
  TCP    0.0.0.0:49666
  TCP    0.0.0.0:49667
  TCP    0.0.0.0:49668
  TCP    0.0.0.0:49676
  TCP    0.0.0.0:49677
  TCP    127.0.0.1:27017
  TCP    127.0.0.1:49671
  TCP    127.0.0.1:49672
  TCP    127.0.0.1:49673
  TCP    127.0.0.1:49674
```

```
student@fedora:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.25
        inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x2
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64
global>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 77  bytes 29443 (28.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 109  bytes 12886 (12.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 20  bytes 2130 (2.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 20  bytes 2130 (2.0 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

⇒ nbtstat -a:

This command helps to solve problems with NETBIOS name resolution

⇒ netstat :

This is a tool used to perform DNS lookups in DNS details.

⇒ nslookup :

Used to perform DNS lookups, nslookup can operate in two modes interactive, non-interactive

Eg: www.google.com

⇒ pathping:

Pathping is a unique to windows and is basically a combination of Tracert commands.

⇒ ping:

Ping is a command to test connectivity nodes.

⇒ Route:

Route command is used to show/manipulate the IP routing table.

```
ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [ME

  -f            Clears the routing tables of all
                used in conjunction with one of
                cleared prior to running the com

  -p            When used with the ADD command,
                boots of the system. By default,
                when the system is restarted. Ig
                which always affect the appropri

  -4            Force using IPv4.

  -6            Force using IPv6.

PS C:\Users\DELL> hostname
HARISH-PC
PS C:\Users\DELL> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::2709:30a3:dc4
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 9:
```

```
sers\DELL>pathping

e: pathping [-g hos
            [-p per
            [-4] [-

ons:
-g host-list      Lo
-h maximum_hops   Ma
-i address        Us
-n                Do
-p period         Wa
-q num_queries    Nu
-w timeout        Wa
-4                Fo
-6                Fo
```

```
PS C:\Users\DELL> nbtstat -a

Displays protocol statistics and current TCP/IP connections using
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table giv
  -A   (Adapter status) Lists the remote machine's name table giv
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] nam
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via
  -R   (Reload)         Purges and reloads the remote cache name
  -S   (Sessions)       Lists sessions table with the destination
  -s   (sessions)       Lists sessions table converting destinati
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and th

  RemoteName    Remote host machine name.
  IP address    Dotted decimal representation of the IP address.
  interval      Redisplays selected statistics, pausing interval s
                between each display. Press Ctrl+C to stop redispl
                statistics.
```

LINUX NETWORKING COMMANDS:

=> ip:

The ip command can show address information manipulate routing, plus display network various devices, interfaces and tunnels

=> ifconfig:

The ifconfig command was a staple in many sys admins tool belt for config and troubleshooting.

=> mtr:

MTR is a program with a CLI that serves as a diagnostic and troubleshooting tool.

=> tcpdump:

The tcpdump command is designed for capturing and displaying packets

=> ping:

Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP)

RESULT:

Basic networking commands in both linux and windows operating system is successfully executed and verified.

```
My traceroute  [v0.95]
st (::1)                                  2026-07-16T1
ode   Restart statistics   Order of fields
                  Packets                      Pings
             Loss%   Snt   Last   Avg  Best
              0.0%   190    0.0   0.1   0.0
```

```
tudent@fedora:~$ ip
sage: ip [ OPTIONS ] OBJECT { COMMAND | help }
      ip [ -force ] -batch filename
here  OBJECT := { address | addrlabel | amt | fou | help | ila |

                link | macsec | maddress | monitor | mptcp | mr

                neighbor | neighbour | netconf | netns | nexthc
                ntbl | route | rule | sr | tap | tcpmetrics |
                token | tunnel | tuntap | vrf | xfrm }
      OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[e
                 -h[uman-r
                 -f[amily]
                 -4 | -6
                 -l[oops]
                 -o[neline
                 -rc[vbuf]
                 -c[olor]
```

```
student@fedora:~$ mtr
student@fedora:~$ su
Password:
root@fedora:/home/student# ps
bash: pseudo: command not fou
root@fedora:/home/student127#
Last metadata expiration chec
.
Package tcpdump-14:4.99.4-2.f
Dependencies resolved.
Nothing to do.
Complete!
root@fedora:/home/student# tc
1.enp0s3 [Up, Running, Connec
2.any (Pseudo-device that cap
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.usbmon2 (Raw USB traffic, bus number 2)
6.usbmon1 (Raw USB traffic, bus number 1)
7.usbmon0 (Raw USB traffic, all USB buses) [none]
8.nflog (Linux netfilter log (NFLOG) interface) [none]
```

```
student@fedora:~# tcpdump -i enp0s3
tcpdump
utput suppressed, use -v[v]... for full protocol decode
s3, link-type EN10MB (Ethernet), snapshot length 262144 byte
P fedora.56515 > dns.google.domain: 26563+ [1au] AAAA? detec
(53)
P fedora.45469 > dns.google.domain: 10457+ [1au] A? detectpo
3)
P dns.google.domain > fedora.45469: 10457 6/0/1 CNAME detect
et., CNAME mozilla.map.fastly.net., A 151.101.1.91, A 151.10
29.91, A 151.101.193.91 (192)
P fedora.59512 > dns.google.domain: 17098+ [1au] A? start.fe
)
P fedora.48751 > dns.google.domain: 15975+ [1au] AAAA? start
(52)
P dns.google.domain > fedora.56515: 26563 6/0/1 CNAME detect
et., CNAME mozilla.map.fastly.net., AAAA 2a04:4e42::347, AAA
, AAAA 2a04:4e42:600::347, AAAA 2a04:4e42:200::347 (240)
P fedora.60875 > dns.google.domain: 12196+ [1au] PTR? 8.8.8.
```

```
student@fedora:~$ ping google.com
PING google.com (142.250.66.14) 56(84) bytes of data.
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=1
e=29.8 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=2
e=37.9 ms
64 bytes from bom07s35-in-f14.1e100.net (142.250.66.14): icmp_seq=3 t
=72.0 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=4
e=35.9 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=5
e=73.9 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=6
e=8.38 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=7
e=91.9 ms
^C
--- google.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6011ms
rtt min/avg/max/mdev = 8.375/49.962/91.899/27.513 ms
```